Status	Finished
Started	Thursday, 10 September 2026, 3:26 PM
Completed	Thursday, 10 September 2026, 3:36 PM
Duration	10 mins 19 secs
Marks	18.00/20.00
Grade	**9.00** out of 10.00 (**90**%)

Question 1

Complete

Mark 4.00 out of 4.00

Select an appropriate technique for each sentence. Note that each key word can be used multiple times.

- This technique is designed for using with fiber optic cable. WDM
- This technique contains two types: synchronous and asynchronous. TDM
- This technique was used by the dial-up modem. FDM
- This technique was used by television broadcasting. FDM

TDM FDM WDM

The correct answer is:
Select an appropriate technique for each sentence. Note that each key word can be used multiple times.

- This technique is designed for using with fiber optic cable. [WDM]
- This technique contains two types: synchronous and asynchronous. [TDM]
- This technique was used by the dial-up modem. [FDM]
- This technique was used by television broadcasting. [FDM]

Question 2

Complete

Mark 2.00 out of 2.00

Six 2 Kbps connections are multiplexed together. Define a data unit is 1 bit.

- The bit duration is 0.5 ms.
- The transmission rate of the aggregated link is 12 Kbps.

Six 2 Kbps connections are multiplexed together. Define a data unit is 1 bit.

- The bit duration = 1/2kbps = 0.5 ms.
- The transmission rate of the aggregated link is = 6 * 2 Kbps = 12 Kbps.

Question 3

Complete

Mark 5.00 out of 6.00

A synchronous TDM combines **four 50-kbps** channels using a time slot of **2 bits**. Calculate the followings:

1. Frame rate (in **kilo**-frames per second) : 25

2. Frame duration (in **milli**second) 0.04

3. Frame duration (in **micro**second) : 40

4. Bit rate of the aggregated link (in **kbps**) : 200

5. Bit interval of the aggregated link (in **milli**second) : 0.005

6. Bit interval of the aggregated link (in **micro**second) : 6.5

Note that, input only the number.

Hint: Slide 38,39

1. Frame rate (in **kilo**-frames per second) = 50 kbps / 2 bits / frame = 25 k frames/sec
2. Frame duration (in **milli**second) = 1/25 k = 0.04 msec
3. Frame duration (in **micro**second) = 0.04 msec * 1,000 = 40 usec
4. Bit rate of the aggregated link (in **kbps**) = 4 * 50 kbps = 200 kbps
5. Bit interval of the aggregated link (in **milli**second) = 1/200 kbps = 0.005 msec
6. Bit interval of the aggregated link (in **micro**second) = 0.005 msec * 1000 = 5 usec

Question 4

Complete

Mark 1.00 out of 1.00

Fill an appropriate answer in the blank.

In statistical or asynchronous TDM , slots are dynamically allocated to improve bandwidth efficiency in the digital transmission.

synchronous TDM | statistical FDM | asynchronous FDM | isochronous TDM

The correct answer is:

Fill an appropriate answer in the blank.

In [statistical or asynchronous TDM], slots are dynamically allocated to improve bandwidth efficiency in the digital transmission.

Question 5

Complete

Mark 0.00 out of 1.00

Which methods is NOT related to circuit switching?

- a. Frequency Division Multiplexing
- b. Wavelength Division Multiplexing
- c. Synchronous Time Division Multiplexing
- d. Asynchronous Time Division Multiplexing

The correct answer is:
Asynchronous Time Division Multiplexing

Question 6

Complete

Mark 6.00 out of 6.00

Given 3 data channels with bit rate 100 kbps, 300 kbps, and 125 kbps using Synchronous TDM to transmit data via a single link.

1. Which channel needs a bit padding? 125kbps
2. What should be the rate of bit padding for that channel? 75 kbps
3. Each frame carries 6 **bits**.
4. The frame rate is therefore 100 **kilo** frames per second since the frame rate is dictated by the minimum-rate channel.
5. The frame duration is 10 **usec**. (microseconds)
6. The bit rate is 600 **kbps**.

Hint: Slide 57

Minimum channel = 100 kbps, but 125 kbps cannot be divided by 100 kbps --> 1.25

So, must do bit padding to have 200 kbps --> bit padding rate = 200-125 = 75 kbps

Finally in one frame, we have 1 slot from 100kbps, 3 slots from 300 kbps, 2 slots from 125 kbps

==> 1 +3 + 2 = 6 slots ==> assume 1 slot = 1 bit (cause no info in the question)

Therefore, each frame contains 6 bits

Frame rate == the minimum channel rate == 100 kilo frames / sec

Frame duration = 1/(100 k) = 10 usec.

The bit rate = 100 kilo frames /sec * 6 bits / frame = 600 kbps